Math 271 Mathematical Methods in Economics, Fall 2026

Tutorial - 3

Topics: Sequences and Subsequences in $\mathbb{R}^n$, Interior and Boundary

1. Let $\{a_n\}_{n\in\mathbb{N}}$ be a convergent sequence in $\mathbb{R}$. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be an injective map and define $b_n = a_{\sigma(n)}$. Prove that $\{b_n\}_{n\in\mathbb{N}}$ is also convergent and that

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} a_n$$

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2. State whether the following statements are True (T) or False (F). Provide rigorous reasons:
   (a) For any sequences $\{x_n\}_{n\in\mathbb{N}}$, $\{y_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}$, if $x_n \to x$ and $d(x_n, y_n) \to 0$, then $y_n \to x$.
   (b) Let $S \subsetneq \mathbb{R}$ be a non-empty finite set, and let $d(x, S) = \min_{z\in S} |x - z|$. Then for all $x_1, x_2 \in \mathbb{R}$:
   $$|d(x_1, S) - d(x_2, S)| \leq |x_1 - x_2|$$

3. Suppose $\{a_n\}_{n\in\mathbb{N}}$ is a sequence of reals with $\{a_{2k}\}_{k\in\mathbb{N}}$ and $\{a_{2k-1}\}_{k\in\mathbb{N}}$ converging to $a_0 \in \mathbb{R}$. Prove that $\{a_n\}_{n\in\mathbb{N}}$ converges to a_0 as well.

4. Let $(n_k)_{k\in\mathbb{N}}$ be any strictly increasing sequence of natural numbers ($n_1 < n_2 < n_3 < \dots$). Prove that for every $N \in \mathbb{N}$, there exists an index k such that $n_j > N$ for all $j \geq k$.

5. Recall the characterization of points for any arbitrary set $A \subseteq \mathbb{R}^n$:
   - $x \in \text{int}(A) \iff \exists\, \epsilon > 0$ such that $B_\epsilon(x) \subseteq A$.
   - $x \in \text{ext}(A) \iff \exists\, \epsilon > 0$ such that $B_\epsilon(x) \subseteq A^c$.
   - $x \in \partial A \iff \forall\, \epsilon > 0$, $B_\epsilon(x) \cap A \neq \emptyset$ and $B_\epsilon(x) \cap A^c \neq \emptyset$.

   Prove the following:
   (a) $\text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B)$
   (b) $\partial(A \cup B) \subseteq \partial A \cup \partial B$